WORKED SOLUTION

STEP III 1992, Question 4

Pure • Orthogonal Trajectories

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q4

The key idea

Write each curve of S_1 as $y = 1 + \dfrac{a}{x-a}$: it is a rectangular hyperbola with asymptotes $x = a$ and $y = 1$, and every one passes through the origin. To find a differential equation for the whole family, differentiate and then use the curve's own equation to get rid of a. Two curves cross at right angles exactly when the product of their gradients is -1, so replacing $\mathrm{d}y/\mathrm{d}x$ by $-1\big/\frac{\mathrm{d}y}{\mathrm{d}x}$ turns the equation for S_1 into the one for S_2, which separates.

Solution

The curves of S_1

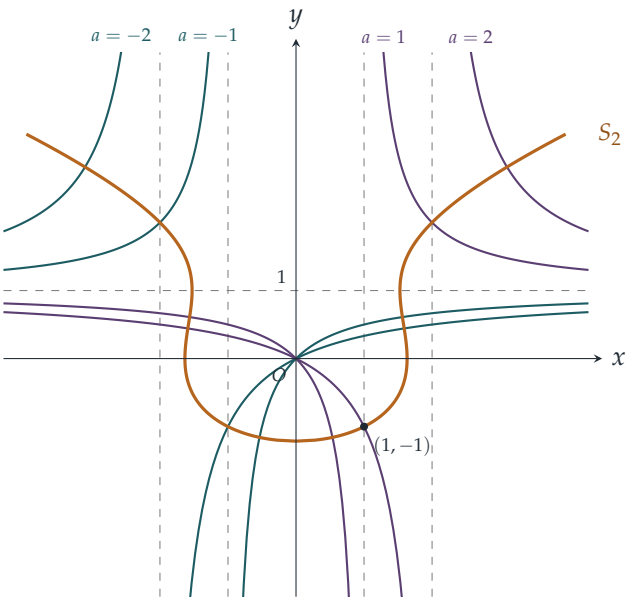
Since $x = (x-a) + a$,

$$y = \frac{x}{x-a} = 1 + \frac{a}{x-a}, \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{a}{(x-a)^2}.$$

So for $a \neq 0$ the curve is a rectangular hyperbola with vertical asymptote $x = a$ and horizontal asymptote $y = 1$, and it passes through the origin.

- If $a > 0$ the gradient is always negative. The branch with $x < a$ comes down from $y = 1$ (as $x \to -\infty$), passes through O and falls to $-\infty$ as $x \to a^-$; the branch with $x > a$ comes down from $+\infty$ towards $y = 1$.
- If $a < 0$ the gradient is always positive. The branch with $x > a$ rises from $-\infty$ through O towards $y = 1$; the branch with $x < a$ lies above $y = 1$.

Replacing x by $-x$ turns $\frac{x}{x-a}$ into $\frac{x}{x+a}$, so the curve for $-a$ is the reflection in the y-axis of the curve for a. A handy point to plot: $y = 3$ when $x = \frac{3}{2}a$.



The dashed lines are the asymptotes. The amber curve is the member of S_2 through $(1,-1)$, whose shape we justify below.

The differential equation for S_1

From $y(x-a)=x$ we get $x-a=\dfrac{x}{y}$ and $a=x-\dfrac{x}{y}=\dfrac{x(y-1)}{y}$. (On a curve of S_1, $y=0$ only at the origin, so away from that point we may divide by y.) Substituting into the gradient,

$$\frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{a}{(x-a)^2}=-\frac{x(y-1)}{y}\cdot\frac{y^2}{x^2}=-\frac{y(y-1)}{x},$$

so every member of S_1 satisfies

$$x\frac{\mathrm{d}y}{\mathrm{d}x}=y(1-y).$$

The differential equation for S_2

Two curves meet at right angles exactly when the product of their gradients at the meeting point is -1. So if p_1 is the gradient of the S_1 curve and p_2 that of the S_2 curve at a common point, then $p_1p_2=-1$, i.e. $p_1=-1/p_2$.

Ask yourself The equation we just found is about S_1. Why does it tell us anything about S_2?

Through each point (x,y) (with $x,y\neq 0$) there passes exactly one curve of S_1, the one with $a=x(y-1)/y$, and its gradient there is $-y(y-1)/x$. A curve of S_2 through the

same point must have gradient $p_2 = -1/p_1$, so on S_2

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{-y(y-1)/x} = \frac{x}{y(y-1)}, \qquad \text{that is,} \qquad x = y(y-1)\frac{\mathrm{d}y}{\mathrm{d}x},$$

as required.

The member of S_2 through $(1, -1)$

The equation separates:

$$\int (y^2 - y)\,\mathrm{d}y = \int x\,\mathrm{d}x \quad \Longrightarrow \quad \frac{y^3}{3} - \frac{y^2}{2} = \frac{x^2}{2} + C.$$

At $(1, -1)$: $-\frac{1}{3} - \frac{1}{2} = \frac{1}{2} + C$, so $C = -\frac{4}{3}$. Multiplying by 6, the curve sketched is

$$3x^2 = 2y^3 - 3y^2 + 8.$$

To sketch it, write $g(y) = 2y^3 - 3y^2 + 8$, so $x = \pm\sqrt{g(y)/3}$ and the curve is symmetric in the y-axis. Since $g'(y) = 6y(y-1)$, g has a local maximum $g(0) = 8$ and a local minimum $g(1) = 7$. Both are positive, so g has just one real zero, between -2 and -1 (as $g(-2) = -20$, $g(-1) = 3$); numerically it is $y_0 \approx -1.214$. So the curve:

- has its lowest point at $(0, y_0)$, where $\mathrm{d}y/\mathrm{d}x = 0$;
- has vertical tangents where $y = 0$ (at $x = \pm\sqrt{8/3} \approx \pm 1.63$) and where $y = 1$ (at $x = \pm\sqrt{7/3} \approx \pm 1.53$), since there $y(y-1) = 0$;
- between $y = 0$ and $y = 1$ bends back slightly towards the axis (for $x > 0$ the gradient $x/(y(y-1))$ is negative there), then grows like $y \approx (\frac{3}{2}x^2)^{1/3}$ for large $|x|$.

As a check on the picture, the S_1 curve through $(1, -1)$ has $-1 = \frac{1}{1-a}$, so $a = 2$, with gradient -2 there; the S_2 gradient is $\frac{1}{(-1)(-2)} = \frac{1}{2}$, and indeed $(-2) \cdot \frac{1}{2} = -1$. The vertical tangent at $y = 1$ also makes sense: there the curve crosses $y = 1$, the $a = 0$ member of S_1, at right angles.

Checks

- At $(1, -1)$: $3(1)^2 = 3$ and $2(-1)^3 - 3(-1)^2 + 8 = 3$. ✓
- The S_2 curve meets the $a = 1$ curve at $(2, 2)$: there the S_1 gradient is $-1/(2-1)^2 = -1$ and the S_2 gradient is $2/(2 \cdot 1) = 1$, product -1.
- Numerically, we found all twelve crossings of the amber curve with the four sketched hyperbolas (three each, e.g. $a = -2$ at $(1.5901, 0.4429)$) and the product of gradients was -1.00000000 at each.

Common slips

- Leaving a in the differential equation: the whole point is to eliminate it using $y = x/(x-a)$ itself.

- Using $p_2 = -p_1$ or $p_2 = 1/p_1$ instead of $p_1 p_2 = -1$.
- Sketching the S_2 curve as a simple U: it has vertical tangents at $y = 0$ and $y = 1$ and a small backward bend between them.

Beyond STEP

The curves of S_2 are the *orthogonal trajectories* of S_1. They appear in physics as field lines crossing equipotentials, or streamlines crossing lines of constant velocity potential. The trick here (eliminate the parameter, then replace y' by $-1/y'$) is the standard method, and the resulting first-order equations are a staple of a first ODE course; see Boyce and DiPrima, *Elementary Differential Equations*.

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